

```
In [8]: import pandas as pd
df = pd.read_csv('customers-100.csv')
print(df)
print(df.head())
print(df.isnull())
df.isnull().sum()
df = pd.read_csv('customers-100.csv', index_col='Customer Id')
print(df)
print(df.iloc[0:5, 1:3])
print(df.loc['053d585Ab6b3159', ['First Name', 'Company']])
print(df.loc['cb8E23e48d22Eae': '2354a0E336A91A1', ['First Name', 'Company']])
```

	Index	Customer Id	First Name	Last Name	\
0	1	DD37Cf93aecA6Dc	Sheryl	Baxter	
1	2	1Ef7b82A4CAAD10	Preston	Lozano	
2	3	6F94879bDAfE5a6	Roy	Berry	
3	4	5Cef8BFA16c5e3c	Linda	Olsen	
4	5	053d585Ab6b3159	Joanna	Bender	
..	...	...	...	...	
95	96	cb8E23e48d22Eae	Karl	Greer	
96	97	CeD220bdAaCfaDf	Lynn	Atkinson	
97	98	28CDbC0dFe4b1Db	Fred	Guerra	
98	99	c23d1D9EE8DEB0A	Yvonne	Farmer	
99	100	2354a0E336A91A1	Clarence	Haynes	

	Company	City	\
0	Rasmussen Group	East Leonard	
1	Vega-Gentry	East Jimmychester	
2	Murillo-Perry	Isabelborough	
3	Dominguez, Mcmillan and Donovan	Bensonview	
4	Martin, Lang and Andrade	West Priscilla	
..	...	...	
95	Carey LLC	East Richard	
96	Ware, Burns and Oneal	New Bradview	
97	Schmitt-Jones	Ortegaland	
98	Fitzgerald-Harrell	Lake Elijahview	
99	Le, Nash and Cross	Judymouth	

	Country	Phone 1	\
0	Chile	229.077.5154	
1	Djibouti	5153435776	
2	Antigua and Barbuda	+1-539-402-0259	
3	Dominican Republic	001-808-617-6467x12895	
4	Slovakia (Slovak Republic)	001-234-203-0635x76146	
..	...	...	
95	Guyana	(188)169-1674x58692	
96	Sri Lanka	+1-846-706-2218	
97	Solomon Islands	+1-753-067-8419x7170	
98	Aruba	(530)311-9786	
99	Honduras	(753)813-6941	

	Phone 2	Email Subscription Da	te \
0	397.884.0519x718	zunigavanessa@smith.info	2020-08-
24			
1	686-620-1820x944	vmata@colon.com	2021-04-
23			
2	(496)978-3969x58947	beckycarr@hogan.com	2020-03-
25			
3	+1-813-324-8756	stanleyblackwell@benson.org	2020-06-
02			
4	001-199-446-3860x3486	colinalvarado@miles.net	2021-04-
17			
..	...	...	
...			
95	001-841-293-3519x614	hhart@jensen.com	2022-01-
30			
96	605.413.3198	vkemp@ferrell.com	2021-07-
10			
97	+1-632-666-7507x92121	swagner@kane.org	2021-09-
18			
98	001-869-452-0943x12424	mccarthystephen@horn-green.biz	2021-08-

11			
99	783.639.1472	colleen91@faulkner.biz	2020-03-
11			

```

Website
0      http://www.stephenson.com/
1      http://www.hobbs.com/
2      http://www.lawrence.com/
3      http://www.good-lyons.com/
4      https://goodwin-ingram.com/
..
95     http://hayes-perez.com/
96     https://novak-allison.com/
97     https://www.ross.com/
98     http://watkins.info/
99     http://www.hatfield-saunders.net/

```

```
[100 rows x 12 columns]
```

	Index	Customer Id	First Name	Last Name
0	1	DD37Cf93aecA6Dc	Sheryl	Baxter
1	2	1Ef7b82A4CAAD10	Preston	Lozano
2	3	6F94879bDAfE5a6	Roy	Berry
3	4	5Cef8BFA16c5e3c	Linda	Olsen
4	5	053d585Ab6b3159	Joanna	Bender

	Company	City
0	Rasmussen Group	East Leonard
1	Vega-Gentry	East Jimmychester
2	Murillo-Perry	Isabelborough
3	Dominguez, Mcmillan and Donovan	Bensonview
4	Martin, Lang and Andrade	West Priscilla

	Country	Phone 1	Phone
2 \			
0	Chile	229.077.5154	397.884.0519x7
18			
1	Djibouti	5153435776	686-620-1820x9
44			
2	Antigua and Barbuda	+1-539-402-0259	(496)978-3969x589
47			
3	Dominican Republic	001-808-617-6467x12895	+1-813-324-87
56			
4	Slovakia (Slovak Republic)	001-234-203-0635x76146	001-199-446-3860x34
86			

ite	Email	Subscription Date	Webs
0	zunigavanessa@smith.info	2020-08-24	http://www.stephenson.c
om/ 1	vmata@colon.com	2021-04-23	http://www.hobbs.c
om/ 2	beckycarr@hogan.com	2020-03-25	http://www.lawrence.c
om/ 3	stanleyblackwell@benson.org	2020-06-02	http://www.good-lyons.c
om/ 4	colinalvarado@miles.net	2021-04-17	https://goodwin-ingram.c

[illegible]

2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False
..	...	...	...	...	...	...	...
95	False	False	False	False	False	False	False
96	False	False	False	False	False	False	False
97	False	False	False	False	False	False	False
98	False	False	False	False	False	False	False
99	False	False	False	False	False	False	False

	Phone 1	Phone 2	Email	Subscription Date	Website
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
..	...	...	...	...	...
95	False	False	False	False	False
96	False	False	False	False	False
97	False	False	False	False	False
98	False	False	False	False	False
99	False	False	False	False	False

[100 rows x 12 columns]

	Index	First Name	Last Name	Company
ny \				
Customer Id				
DD37Cf93aecA6Dc	1	Sheryl	Baxter	Rasmussen Group
1Ef7b82A4CAAD10	2	Preston	Lozano	Vega-Gentry
6F94879bDAfE5a6	3	Roy	Berry	Murillo-Perry
5Cef8BFA16c5e3c	4	Linda	Olsen	Dominguez, Mcmillan and Donovan
053d585Ab6b3159	5	Joanna	Bender	Martin, Lang and Andrade
...	...	...	...	
...				
cb8E23e48d22Eae	96	Karl	Greer	Carey LLC
CeD220bdAaCfaDf	97	Lynn	Atkinson	Ware, Burns and Oneal
28CDbC0dFe4b1Db	98	Fred	Guerra	Schmitt-Jones
c23d1D9EE8DEB0A	99	Yvonne	Farmer	Fitzgerald-Harrell
2354a0E336A91A1	100	Clarence	Haynes	Le, Nash and Cross

	City	Country \
Customer Id		
DD37Cf93aecA6Dc	East Leonard	Chile
1Ef7b82A4CAAD10	East Jimmychester	Djibouti
6F94879bDAfE5a6	Isabelborough	Antigua and Barbuda
5Cef8BFA16c5e3c	Bensonview	Dominican Republic
053d585Ab6b3159	West Priscilla	Slovakia (Slovak Republic)
...	...	...
cb8E23e48d22Eae	East Richard	Guyana
CeD220bdAaCfaDf	New Bradview	Sri Lanka

28CDbC0dFe4b1Db c23d1D9EE8DEB0A 2354a0E336A91A1	Ortegaland Lake Elijahview Judymouth	Solomon Islands Aruba Honduras
	Phone 1	Phone 2 \
Customer Id		
DD37Cf93aecA6Dc	229.077.5154	397.884.0519x718
1Ef7b82A4CAAD10	5153435776	686-620-1820x944
6F94879bDAfE5a6	+1-539-402-0259	(496)978-3969x58947
5Cef8BFA16c5e3c	001-808-617-6467x12895	+1-813-324-8756
053d585Ab6b3159	001-234-203-0635x76146	001-199-446-3860x3486
...	...	...
cb8E23e48d22Eae	(188)169-1674x58692	001-841-293-3519x614
CeD220bdAaCfaDf	+1-846-706-2218	605.413.3198
28CDbC0dFe4b1Db	+1-753-067-8419x7170	+1-632-666-7507x92121
c23d1D9EE8DEB0A	(530)311-9786	001-869-452-0943x12424
2354a0E336A91A1	(753)813-6941	783.639.1472
	Email Subscription Date \	
Customer Id		
DD37Cf93aecA6Dc	zunigavanessa@smith.info	2020-08-24
1Ef7b82A4CAAD10	vmata@colon.com	2021-04-23
6F94879bDAfE5a6	beckycarr@hogan.com	2020-03-25
5Cef8BFA16c5e3c	stanleyblackwell@benson.org	2020-06-02
053d585Ab6b3159	colinalvarado@miles.net	2021-04-17
...	...	...
cb8E23e48d22Eae	hhart@jensen.com	2022-01-30
CeD220bdAaCfaDf	vkemp@ferrell.com	2021-07-10
28CDbC0dFe4b1Db	swagner@kane.org	2021-09-18
c23d1D9EE8DEB0A	mccarthystephen@horn-green.biz	2021-08-11
2354a0E336A91A1	colleen91@faulkner.biz	2020-03-11
	Website	
Customer Id		
DD37Cf93aecA6Dc	http://www.stephenson.com/	
1Ef7b82A4CAAD10	http://www.hobbs.com/	
6F94879bDAfE5a6	http://www.lawrence.com/	
5Cef8BFA16c5e3c	http://www.good-lyons.com/	
053d585Ab6b3159	https://goodwin-ingram.com/	
...	...	
cb8E23e48d22Eae	http://hayes-perez.com/	
CeD220bdAaCfaDf	https://novak-allison.com/	
28CDbC0dFe4b1Db	https://www.ross.com/	
c23d1D9EE8DEB0A	http://watkins.info/	
2354a0E336A91A1	http://www.hatfield-saunders.net/	

[100 rows x 11 columns]

	First Name	Last Name
Customer Id		
DD37Cf93aecA6Dc	Sheryl	Baxter
1Ef7b82A4CAAD10	Preston	Lozano
6F94879bDAfE5a6	Roy	Berry
5Cef8BFA16c5e3c	Linda	Olsen
053d585Ab6b3159	Joanna	Bender
First Name		Joanna
Company		Martin, Lang and Andrade
Name: 053d585Ab6b3159, dtype: object		
	First Name	Company
Customer Id		
cb8E23e48d22Eae	Karl	Carey LLC

CeD220bdAaCfaDf	Lynn	Ware, Burns and Oneal
28CDbC0dFe4b1Db	Fred	Schmitt-Jones
c23d1D9EE8DEB0A	Yvonne	Fitzgerald-Harrell
2354a0E336A91A1	Clarence	Le, Nash and Cross

Interpolate()

```
In [9]: #interpolate()
#Estimate missing values.
import pandas as pd
sample = pd.DataFrame({
    'Age': [20, 24, 25, None, 40, 35]
})
sample['Age'] = sample['Age'].interpolate()
print(sample)
```

	Age
0	20.0
1	24.0
2	25.0
3	32.5
4	40.0
5	35.0

In []: